2026-04-24

# HydraR: Agentic Infrastructure for Reproducible Scientific Research

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## Executive Summary

Large Language Models (LLMs) are catalyzing a paradigm shift in scientific computing, fundamentally transforming how research is performed. This transformation is particularly evident in the field of bioinformatics, where LLMs have evolved beyond simple chatbots into autonomous "agents" that can plan and execute complex tasks that previously demanded computational expertise, and powerful multi-agent systems, where a team of AI agents can autonomously collaborate with one another to tackle multi-faceted problems. However, the current R ecosystem lacks a robust framework that can support the development and use of agentic systems while safeguarding the integrity of R-based research. Because of this, multi-agent systems poses a significant risk to the reproducibility of R-based scientific research. In this project, we aim to address this critical need by delivering a robust framework for traceable, stateful, and secure agent orchestration. To accomplish this, we will leverage the existing functionality of HydraR, an R package for the scientific reproducibility of agentic systems. Specifically, we aim to develop a comprehensive memory and state management system for logging ephemeral agent reasoning and agent interactions into auditable records for peer review and regulatory compliance, and an environment management system to ensure agents can iterate and fail safely without risking the integrity of sensitive research environments. Thus, by serving as the central hub for traceable, stateful and secure agentic orchestration, HydraR will empower computational biologists and the broader R community to adopt agentic systems with confidence, and build sophisticated, transparent and reproducible AI-driven workflows that uphold the rigor and integrity of R-based scientific research.

### Expected Outcomes

1. **HydraR R Package**: A comprehensive toolkit for building stateful agentic graphs.
2. **Resilience Layer**: A native checkpointing system to minimize failure costs.
3. **Audit Logs**: Structured, SQL-queryable records of all agent reasoning and tool interactions.
4. **Community Resources**: Documentation and vignettes demonstrating reproducible workflows for the R Consortium.

### Budget

Total requested budget: **$14,400 USD**. We request funding for expert software developers to work on this project over a 7-week period (approximately 245 hours).

# Signatories

## Project team

- **Chi Nam (Ignatius) Pang** (ORCID: 0000-0001-9703-5741). **Co-Primary Investigator**. Senior Bioinformatician at Macquarie University / Australian Proteome Analysis Facility (APAF). Expert in R package development, proteomics pipelines, and reproducibility standards.
- **Aidan Tay** (ORCID: 0000-0003-1315-4896). **Co-Primary Investigator (Co-CI)**. Bioinformatician at APAF. Architect of the HydraR R6 DAG engine, Git Worktree isolation logic, and the custom state-hashing persistence Layer.

## Contributors

- **APAF Bioinformatics Team**: Provided the testing ground for initial prototypes in high-throughput proteomics and metabolomics research.
- **Antigravity AI**: Agentic assistant used during the rapid prototyping and documentation hardening phase of the project (as disclosed in `agents.md`).

## Consulted

- **rOpenSci Editorial Board**: Initial discussions regarding HydraR's role in the AI-R ecosystem.
- **Bioplatforms Australia (BPA)**: Consulted on the impact of agentic orchestration for national-scale omics infrastructure.

# The Problem

Large Language Models (LLMs) are catalyzing a paradigm shift in scientific computing, fundamentally transforming how research is performed. This transformation is particularly evident in the field of bioinformatics, where LLMs have evolved beyond simple chatbots into autonomous "agents" that can plan and execute complex tasks previously requiring computational expertise, and into powerful multi-agent systems, where a team of AI agents can autonomously collaborate to tackle multifaceted problems. By democratizing access to these sophisticated multi-agent systems, this automation not only accelerates biological discovery but also empowers the broader scientific community to harness high-performance workflows once restricted to deep domain specialists.

Yet despite offering many great benefits, the current R ecosystem lacks the necessary framework for traceable, stateful, and secure agent orchestration. Existing R packages such as `ellmer` and `mall` provide excellent interfaces to LLM providers to send prompts and parse responses but fail to record the entire decision-making process of an agent, making the "journey" ephemeral and untraceable. Other packages such as `LLMAgentR` and `puppeteeR` enable agent orchestration, but lack the infrastructure for state management, making it impossible for multi-agent systems to meet the rigorous reproducibility standards of modern bioinformatics. Meanwhile, advanced frameworks such as `LangGraph` and `AutoGen` in the Python ecosystem lack the necessary features for auditable agent orchestration, rendering them unsuitable for R-based research via Python-based bridges (i.e.,

`reticulate`). Moreover, granting autonomous agents unchecked access to research environments can be dangerous as their actions can inadvertently lead to file modifications, deletions, or corruption of the working directory. Thus, to safeguard the integrity of R-based research, a robust framework must exist to support the development and reproducible use of agentic systems.

To address this critical need, this project will build on HydraR, an existing R package dedicated to the scientific reproducibility of agentic systems. To accomplish this, we aim to develop a comprehensive memory and state management system for logging ephemeral agent reasoning and agent interactions into auditable records for peer review and regulatory compliance. We will also develop an environment management system to ensure agents can iterate and fail safely without risking the integrity of sensitive research environments. Thus, by serving as the central hub for traceable, stateful and secure agentic orchestration, HydraR will empower computational biologists and the broader R community to adopt agentic systems with confidence, and build sophisticated, transparent and reproducible AI-driven workflows that uphold the rigor and integrity of R-based scientific research.

# The Proposal

## Overview

The current R ecosystem lacks the necessary framework for traceable, stateful and secure agent orchestration. To address this critical gap, this project will build on HydraR, an existing R package dedicated to the scientific reproducibility of agentic systems. By leveraging the existing functionality of HydraR, this project aims to deliver a robust framework for traceable, stateful and secure agent orchestration. In doing so, HydraR will empower computational biologists and the broader R community to adopt agentic systems with confidence, and build sophisticated, transparent and reproducible AI-driven workflows that uphold the rigor and integrity of R-based scientific research.

## Details

This project aims to address the critical need for traceable, stateful, and secure agent orchestration in the R ecosystem. To do this, we will extend on HydraR, an existing R package dedicated to the scientific reproducibility of agentic systems. In particular, we aim to address this gap through two objectives:

## Objective 1: Memory and State Management

**Aim**: To develop a comprehensive memory and state management system for logging ephemeral agent reasoning and agent interactions into auditable records for peer review and regulatory compliance.

**Rationale**: Agent memory is often ephermeral and untraceable, making it difficult to capture and audit the entire "journey" of an agentic workflow. Currently, HydraR uses DuckDB to maintain a central memory system that continuously saves workflow checkpoints, ensuring a complete, verifiable record of all AI interactions and data changes. Despite this, HydraR does not support automated retries for failed tasks, execution caching and state retreival for skipping duplicated tasks and resource optimization, or agent validation for workflow resilience, which are essential for production-grade bioinformatic pipelines.

**Approach**: To develop a comprehensive memory and state management system, we will introduce several enhancements to HydraR:

- **Restart Checkpointing**: We will implement a system to automatically serialize agents, environment, local variables, and their current reasoning state. Thus, in the event of a failure (API timeout, resource exhaustion, or logic error), HydraR can reconstitute the exact state of the R session, allowing for automated retries without restarting the entire pipeline.

- **Execution Caching and State Retrieval**: We will implement a content-addressable storage (CAS) model for all agent-generated artifacts. By hashing output files and R objects, and linking them to specific states and reasoning paths, HydraR can maintain a complete audit trail of how each artifact was generated, allowing the system to "skip" previously executed tasks and "resume" from any point in the workflow.

- **Agent Validation and Recovery**: We will implement a system to automatically validate agent outputs. Thus, if the output of an agent fails to validate (e.g., malformed code or incorrect data schema), the orchestration engine can dynamically inject a "repair" to diagnose and rectify the error, ensuring the main research flow continues safely and autonomously.

### Objective 2: Environment Management

**Aim**: To develop an environment management system to ensure agents can iterate and fail safely without risking the integrity of sensitive research environments.

**Rationale**: Granting autonomous agents unchecked access to sensitive research environments can be dangerous as their actions can inadvertently lead to file modifications, deletions, or corruption of the working directory. Unrestricted access to the working directory also makes it difficult for parallelised agents to work on the same task without overwriting each other's work, which is a major hurdle for scaling agentic workflows. One way to mitigate this risk is to enforce filesystem and runtime isolation using containerisation. However, HydraR currently lacks the capabilities to safely and securely orchestrate agentic workflows, with integration with filesystem and runtime isolation tools such as Git Worktree, Docker and Singularity/Apptainer for High-Performance Computing (HPC) environments.

**Approach**: To develop a comprehensive environment management system, we will introduce several new features to HydraR:

- **Git Worktrees**: We will implement a Git worktree layer to prevent agents from modifying the filesystem at the same time. By assigning every task to a unique worktree, HydraR can capture and track all file changes made by the agent. This prevents accidental corruption of the main research branch as well as an auditable record of every file modification.

- **Docker Integration**: We will implement a Docker layer where each agent execution environment is isolated within a container. Every agent can operate in its own ephemeral environment, allowing the orchestrator to spin up and shut down environments as needed. This will also ensure that the agentic workflows are reproducible, as each agent can be executed in the same environment.

- **Singularity and Apptainer for HPC**: Recognizing that many bioinformatics workflows are hosted on institutional HPC clusters, where Docker may be restricted, we will provide first-class support for Singularity and Apptainer. We will tailor this support to standard HPC structures (e.g., SLURM integration and shared filesystem mounts), ensuring that the agentic infrastructure respects the security and resource policies of high-performance computing centers.

- **Persistent State Tracking via Mounted Worktrees**: To bridge the gap between container volatility and workflow statefulness, HydraR will manage Git worktree logic within persistent Docker mounts. By mounting the worktree directory from the host to the container, we ensure that the agent's filesystem state is preserved across container restarts. Combined with our `on_failure` checkpointing, this allows an agent to resume from the exact filesystem state it was in before a crash, while maintaining a versioned history of its progress via Git.

# Timeline

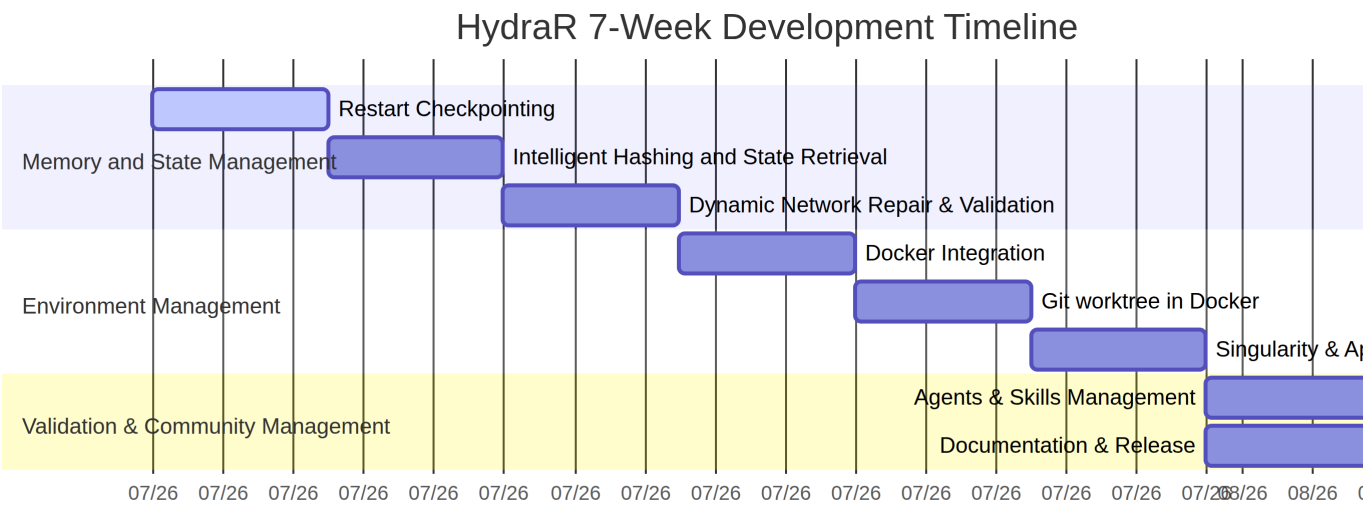
## Start-up Phase

- **Reporting Framework**: Establish a bi-weekly internal review process to track milestone progress.

## Technical Delivery

The proposed work will be delivered over a 7-week period and involve approximately 245 hours of expert developer time. The work is structured into three phases:

- **Phase 1: Memory and State Management (Weeks 1-3)**:
  - Development of automated state persistence and restart checkpointing.
  - Development of the serialization engine.
  - Development of dynamic sub-graph injection for self-correcting flows.
  - Development of the validation and pruning layers for network preparedness.
- **Phase 2: Environment Management (Weeks 4-6)**:
  - Development of Docker Integration
  - Development of Git worktree for state tracking in containers
  - Development of compatibility mapping for Singularity and Apptainer
- **Phase 3: Validation & Community Management (Weeks 6-7)**:
  - Finalization of Agents and Skills Management tools into lifecycle modules.
  - Documentation, bug fixing, and release processing.




## Other Aspects

- **Hosting**: Code will be hosted on GitHub under the `APAF-bioinformatics` organization.
- **Conference**: We will attend and present HydraR at relevant conferences to engage with the bioinformatics community.
- **Social media**: We will utilize social media to engage with the bioinformatics community.
- **Publicity**: We will share quarterly updates on the R Consortium blog.
- **Community Engagement**: We will share undergo the rOpenSci peer-review process and transition to a community-maintained model.

## Budget & Funding Plan

Total requested budget: **$14,400 USD**. We request funding for expert software developers to work on this project over a 7-week period (approximately 245 hours).

| Milestone | Description | Estimated Hours | Funding |
|---|---|---|---|
| **Memory and State Management** | [Details of Milestone 1] | 140 | $5,760 |
| **Environment Management** | [Details of Milestone 2] | 140 | $5,760 |
| **Validation & Community Management** | [Details of Milestone 3] | 80 | $2,880 |
| **Total** | | **360** | **$14,400** |

| | R ConsortiumCash (USD) | Macquarie UniversityIn-kind |
|---|---|---|
| Aidan Tay (1.0 FTE) | $14,400 | - |
| Chi Nam (Ignatius) Pang (0.2 FTE) | - | $3,431 |
| Total | **$14,400** | **$3,431** |

# Success

The success of the HydraR infrastructure project will be measured by its adoption, technical stability, and contribution to the R ecosystem's AI readiness.

## Success Metrics

- **Package Adoption**: At least 3 existing R/Bioconductor packages demonstrating "Skill Node" integration via the HydraR manifest protocol.
- **Reliability Benchmark**: A published reproducible benchmark dataset showing a >20% improvement in agent task completion rates when using HydraR state management vs. stateless alternatives.
- **Community Engagement**: Successful completion of the rOpenSci peer-review process and transition to a community-maintained model.
- **CRAN/Bioconductor Presence**: Maintain the 0-error/warning/note status across all major R versions.

## Definition of Done

The project will be considered successful upon the delivery of:

- A stable CRAN-ready (or Bioconductor-ready) R package named `HydraR`.
- A fully functional "Checkpoint & Resume" engine demonstrated in a multi-agent bioinformatics workflow.
- The capability to perform SQL queries over agent reasoning history via an integrated DuckDB backend.
- Comprehensive documentation, including at least three "reproducible case studies" (vignettes) showcasing traceable agent orchestration.

## Measuring success

We will use the following markers to track our progress:

- **Week 2**: Completion of the `duckdb` backend and successful hashing of R objects in a session.
- **Week 4**: Passing unit tests for state serialization and automated recovery from simulated API failures.
- **Week 6**: Successful community beta-test with at least 5 bioinformatics researchers.
- **Week 8**: Final project report and publication of the demonstration blog post on the R Consortium website.

## Future work

HydraR is backed by **APAF Bioinformatics** (at Macquarie University), ensuring long-term institutional support and alignment with national-scale bioinformatics infrastructure (NCRIS).

- **Commercial & Institutional Support**: Ongoing maintenance is built into the APAF digital transformation roadmap (2026-2030).
- **Open Source Community**: By providing an interoperable manifest protocol, we encourage a "plugin" ecosystem where community members can contribute drivers and skill nodes without needing deep knowledge of the core DAG engine, ensuring the framework outlives its original funding period.

HydraR is designed to be the foundation for an "Autopoietic Agentic Ecosystem" in R. Future development directions include:

- **Decentralized Orchestration**: Extending the infrastructure to support peer-to-peer agent coordination across distributed R nodes.
- **Advanced Containerization**: Tighter integration with Docker/Singularity for "Secure Execution Enclaves" where agents can test code in isolated environments.
- **Agentic Benchmarking**: A standardized suite for benchmarking LLM performance on bioinformatics-specific R tasks using the HydraR traceability layer.